ToPP! Topology of Pulsar Profiles

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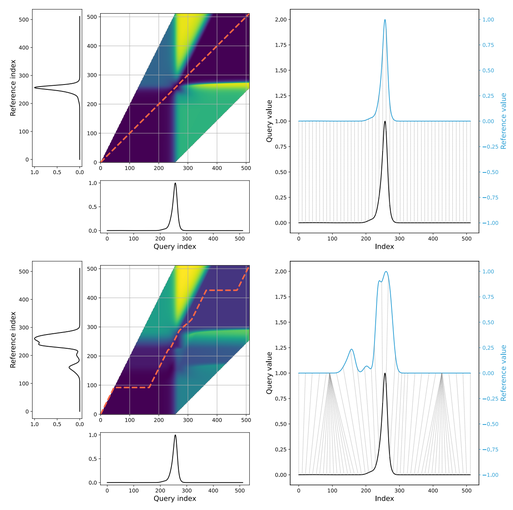
article with J. van Leeuwen & Y. Maan

Comparing profile's Stokes I morphology



$w = 0$ if identical

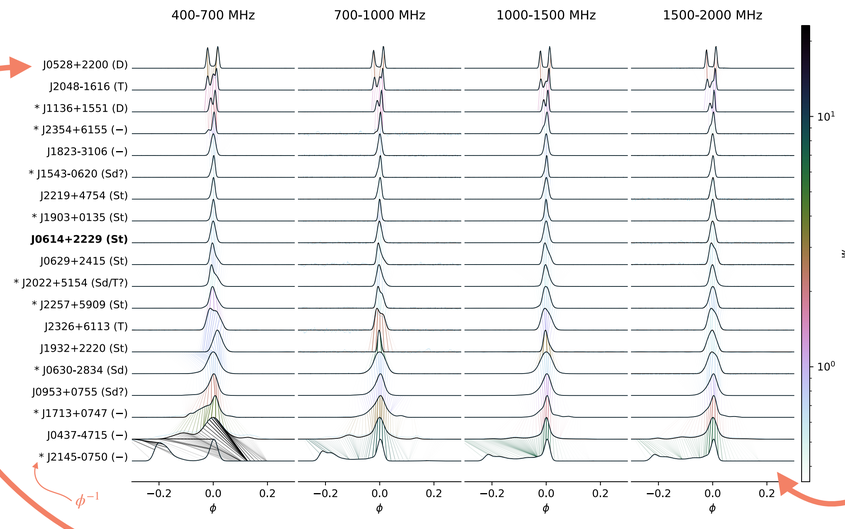
$w > 0$ if different

A **pulsar average profile** (intensity vs rotational phase) is its signature — essential tool for pulsar timing

Average profiles are (mostly) time-stable carrying info about emission regions and geometry

How do 90 pulsars relate?

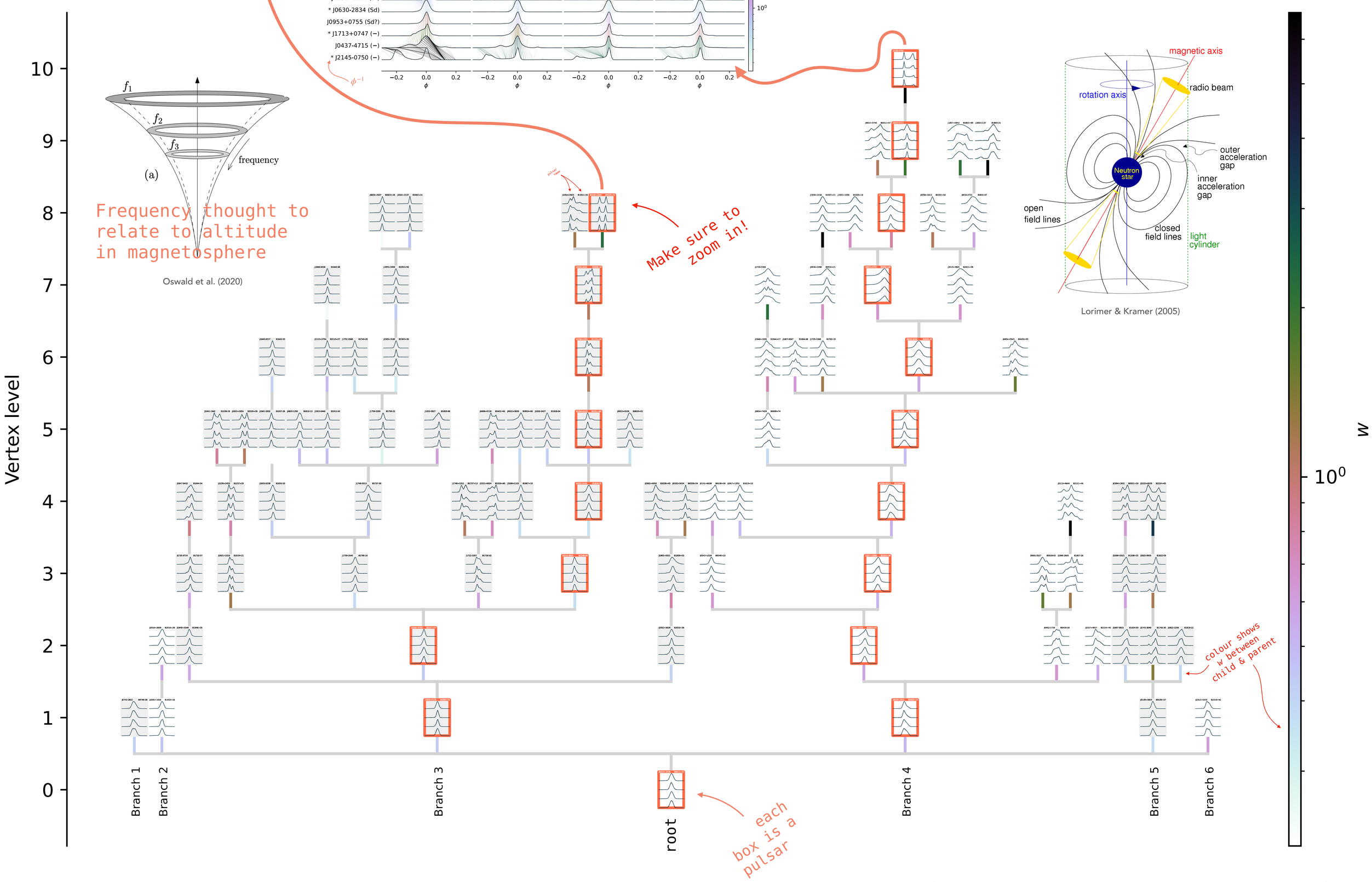
We use graph theory to investigate the European Pulsar Network database.



Sequencing highlights morphology evolution across pulsars

Minimum Spanning Tree on w

(organizes pulsars by spectro-temporal morphology)



Frequency thought to relate to altitude in magnetosphere

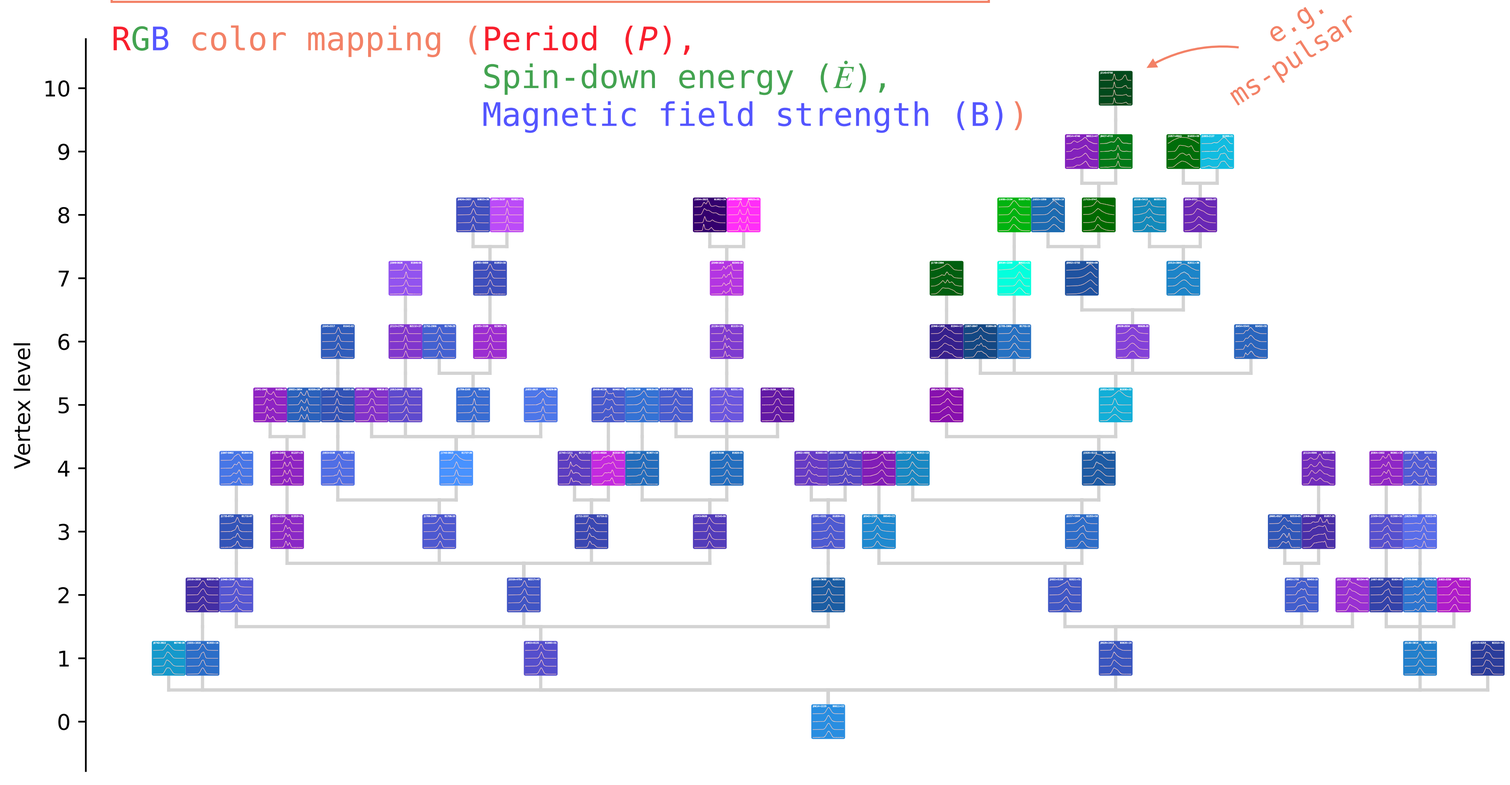
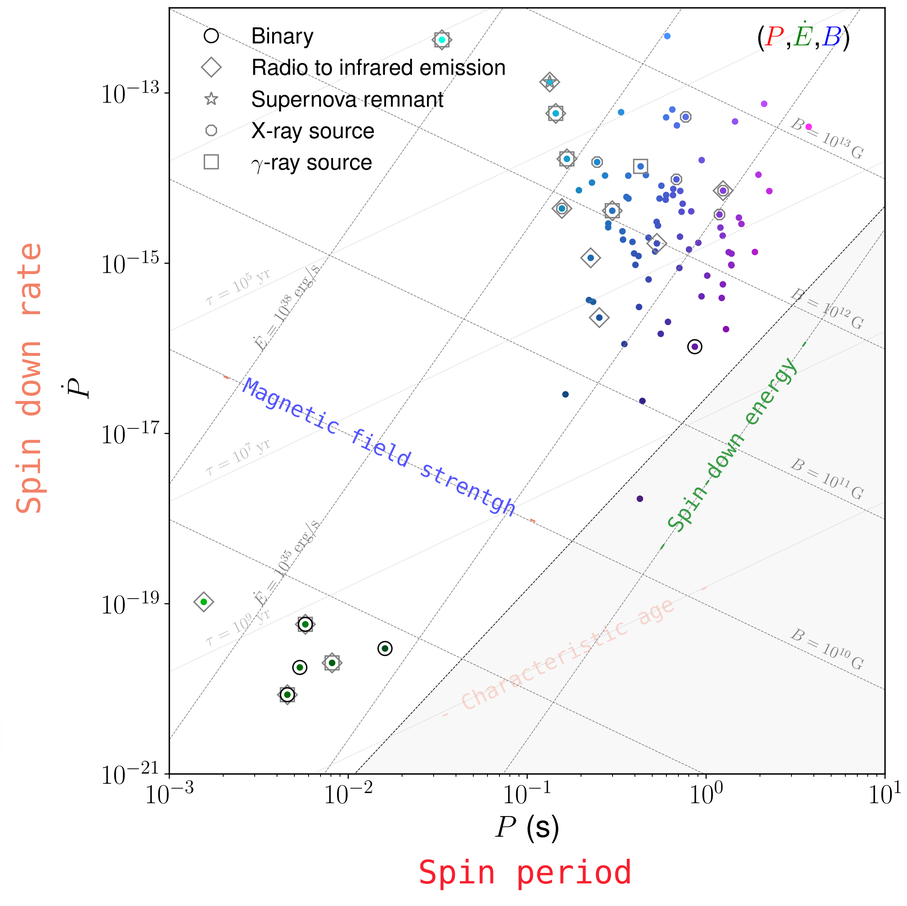
Oswald et al. (2020)

Lorimer & Kramer (2005)

Make sure to zoom in!

colour shows w between child & parent

each box is a pulsar

We can relate graph regions to physical parameters

RGB color mapping (Period (P), Spin-down energy ($\dot{E}$), Magnetic field strength (B))